

1937

5. (a) Describe various types of placenta on a histological basis. 10
(b) Add a note on the functions of placenta. 5
6. Write short notes on any **three** of the following :
5×3=15
- (i) Extraembryonic membranes in chick
 - (ii) Fate Maps
 - (iii) In vitro fertilization
 - (iv) Hormonal disrupters as teratogens
 - (v) Patterns of cleavage

[This question paper contains 4 printed pages]

Your Roll No.

Sl. No. of Q. Paper

Unique Paper Code

Name of the Paper

Name of the Course

Semester

Time : 2 Hours

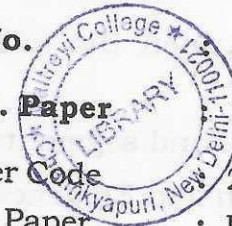
Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) Attempt any **four** questions in **all**. Question **NO.1** is compulsory.
- (c) Attempt **all** the parts of question together.
- (d) Draw-well labelled diagrams wherever necessary.

1. (a) Define (any **four**) :

- (i) Perivitelline space
- (ii) Blastema
- (iii) Cortical granules

P.T.O.



03.06.2026 (M)

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: DSC : Development
Biology

: **B.Sc.(Hons.)**
Zoology, NEP, UGCF

: IV

Maximum Marks : 60

1×4=4

- (iv) Senescence
- (v) Polar body
- (b) Differentiate between the following (any **two**) : $2 \times 2 = 4$
- Subgerminal and segmentation cavity
 - Preformation and Epigenesis
 - Oligolecithal and Polylecithal eggs
- (c) Name the germ layers from which each of the following is derived. $0.5 \times 6 = 3$
- Lens of Eye
 - Adrenal Medulla
 - Skin
 - Bone
 - Notochord
 - Pituitary
- (d) Expand the following abbreviations: $0.5 \times 4 = 2$
- GVBD
 - IMZ
 - ESC
 - DES

- (e) State the major contributions of the following scientists : $1 \times 2 = 2$
- W. Vogt
 - Hilde Mangold
2. (a) Explain in detail the growth of oocyte during oogenesis. 8
- (b) Describe the role of acrosome reaction during fertilization in mammals. 7
3. (a) Define Metamorphosis. Briefly describe hormonal regulation of metamorphosis in insects. 8
- (b) Explain the principle and procedure of Amniocentesis. Discuss its significance in the prenatal diagnosis. 7
4. (a) Describe various morphogenetic movements with reference to gastrulation in frog. 10
- (b) Compare the blastopore of frog to primitive streak of chick. 5